



PROJECT-BASED LEARNING

PHASE-I EVALUATION

Smart Campus Navigation System

Team ID: DSCPP-III-2026-T013

Department of Computer Science & Engineering
Graphic Era (Deemed to be University), Dehradun
Academic Session 2026-27



Team & Mentor Details

Team Information

Team ID: DSCPP-III-2026-T013
Semester: 3rd
Section: BTECH
Domain: CSE

Team Members

1. *Vanshika Aggarwal – 2510011410*
2. *Suvidha Setia – 2510370306*
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4. *Khushi Taneja – 2510379518*

Mentor

Ms. Pratibha Waldai

Interactions completed:4



Problem Statement & Motivation

Problem Statement

Finding a particular place inside a large college campus can be confusing, especially for new students and visitors. Multiple academic blocks, labs, offices, hostels, and other facilities may be located at different parts of the campus. General maps may not show detailed internal campus routes.

Motivation

- ▶ *Difficulty finding unfamiliar locations*
- ▶ *Dependence on others for directions*
- ▶ *Lack of detailed internal routes*

Target Users / Stakeholders

- ▶ *Students*
- ▶ *Faculty*
- ▶ *Visitors*



Objectives

Project Objectives

1. *Develop a campus navigation system using DSA.*
2. *Represent campus locations as a graph.*
3. *Implement the graph using C++.*
4. *Find efficient routes between locations.*
5. *Apply Dijkstra's Algorithm.*
6. *Test different routes.*

Key Objective: *Find an efficient route between two selected campus locations.*



Proposed Solution

Proposed Approach

1. *Input: Select source and destination.*
2. *Data Collection: Identify campus locations and paths.*
3. *Processing: Represent campus as a graph.*
4. *Algorithm: Apply Dijkstra's Algorithm.*
5. *Output: Display the efficient route.*

Key Features

- ▶ *Campus location selection*
- ▶ *Graph-based representation*
- ▶ *Shortest-path calculation*
- ▶ *Route finding*
- ▶ *Multiple route testing*



Integration of PBL Course Concepts

Course	Concepts Applied	Project Module
Data Structures in C	Graphs, Adjacency List, Graph Traversal	Campus Graph Representation
OOPs with C++	Classes, Objects, STL	System Implementation

Requirement: The proposed solution addresses the problem by using graph-based shortest-path algorithms to provide an efficient route between selected campus locations.



System Architecture / Workflow



Major Components

- ▶ Location Input Module
- ▶ Campus Graph Module
- ▶ Shortest Path Module

Data / Control Flow

- ▶ User selects source and destination
- ▶ Graph is processed using Dijkstra's Algorithm
- ▶ Efficient route is displayed



Technology Stack & Methodology

Technology Stack

- ▶ Programming: *C / C++ / DS*
- ▶ Database: *Graph, File-based storage*
- ▶ Tools / Framework: *C++*
- ▶ IDE: *Visual Studio Code*
- ▶ Version Control: *Git / GitHub*

Development Methodology

1. Requirement Analysis
2. System Design
3. Implementation
4. Testing
5. Evaluation

Justify the selection of technologies and methodology.



Initial Progress & Team Contribution

Progress Achieved

- ▶ *Problem identified*
- ▶ *Requirements identified*
- ▶ *Campus locations and pathways planned*
- ▶ *Graph-based approach selected*
- ▶ *Algorithm selected: Dijkstra's Algorithm*

Individual Contribution

Member	Role	Contribution
F	Lead	System Designer
ML1	Developer	Graph & Algorithm Developer
H	DB/Testing	Data & Testing Engineer
ML2	Developer/Docs	Documentation & Research



Roadmap, Expected Outcomes & References

Roadmap

1. Phase-I: Proposal & Design
2. Phase-II: Development
3. Phase-III: Final Implementation

Expected Outcomes

- ▶ *Working prototype / system*
- ▶ *Measurable results*
- ▶ *Practical applicability*
- ▶ *Future scalability*

References

1. DOI:<https://doi.org/10.55041/ijsmv.v2i5.089>
2. DOI:<https://doi.org/10.18488/79.v12i1.4656>
3. <https://doi.org/10.34739/si.2023.29.06>
4. <https://doi.org/10.1186/s13673-020-00222-0>

Thank You
Questions &
Suggestions